ASEAN
GEOAI FUSION 2026

Geospatial Data Fusion

The Pipeline from Raw Sensors to Actionable Intelligence

Dr. Fathoni Usman, M.Eng., Eng.Tech., IPM, ASEAN Eng.
UNITEN



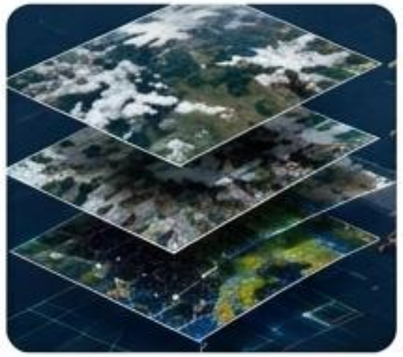
ASEAN
GEOAI FUSION 2026

The Geospatial Pipeline: From Raw Data to Intelligence

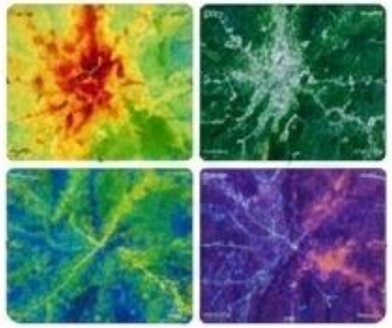
RAW DATA



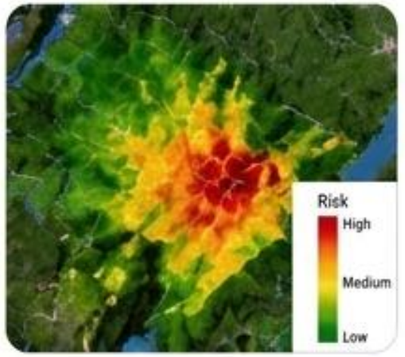
INFORMATION



FEATURES & INDICATORS



PREDICTIONS



ACTIONABLE DECISIONS



Stage 1: Raw Data

Unprocessed observations, large volume, no context.

Stage 2: Information

Cleaned, georeferenced, structured datasets.

Stage 3: Features & Indicators

Derived information, extracted patterns, AI-ready inputs.

Stage 4: Predictions

Risk/susceptibility maps, probability estimates, scenarios.

Stage 5: Actionable Decisions

Resource optimization, policy policy planning, and real-world impact.

The Imperative for Data Fusion

The Problem

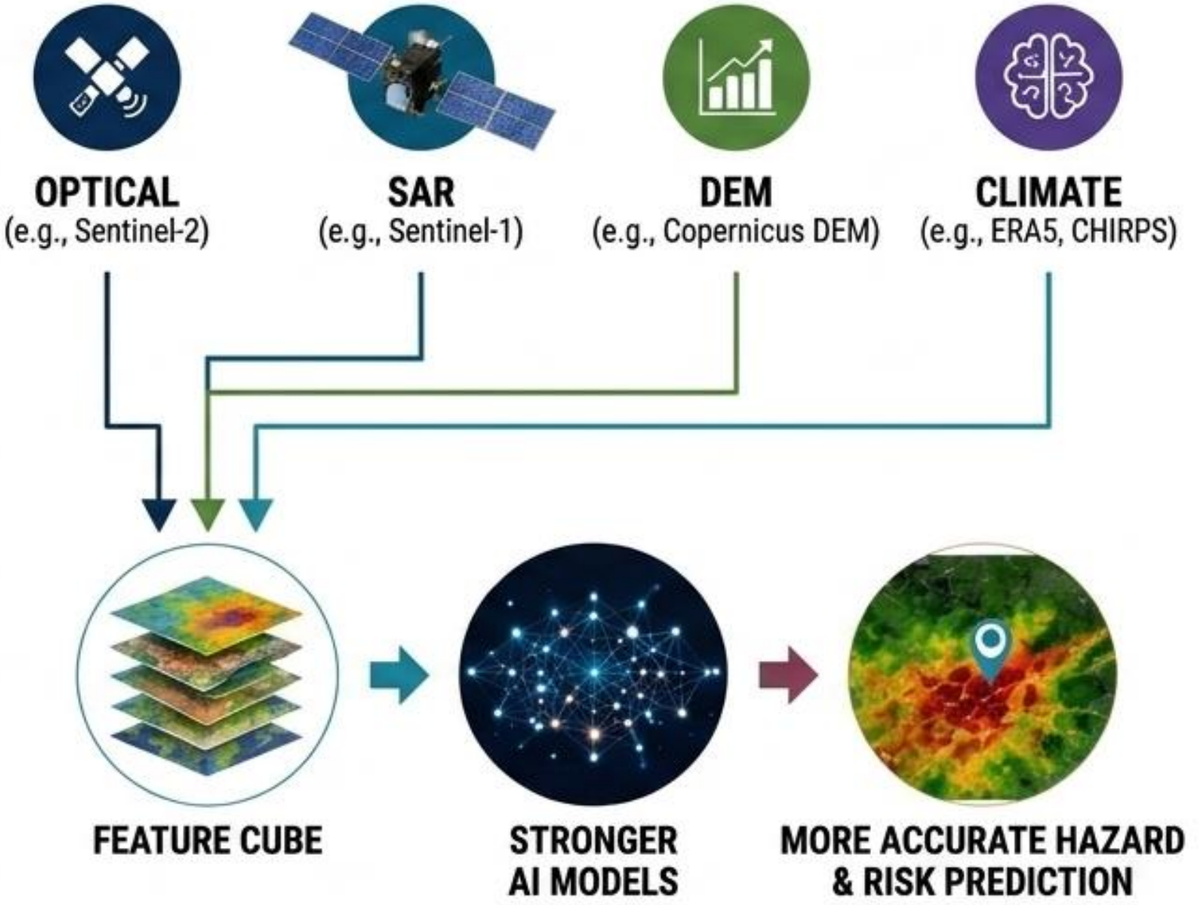
No single sensor sees the whole picture. Optical imagery misses cloud-covered areas, radar struggles with fine detail, and ground sensors only cover discrete points.

The Operational Cost

A missed signal, a flood, an outage, a land-use change, carries severe operational and economic consequences.

The Solution

Fusion reduces blind spots, cancels out random noise through redundancy, and produces a more complete and statistically reliable picture than any one input could deliver on its own.

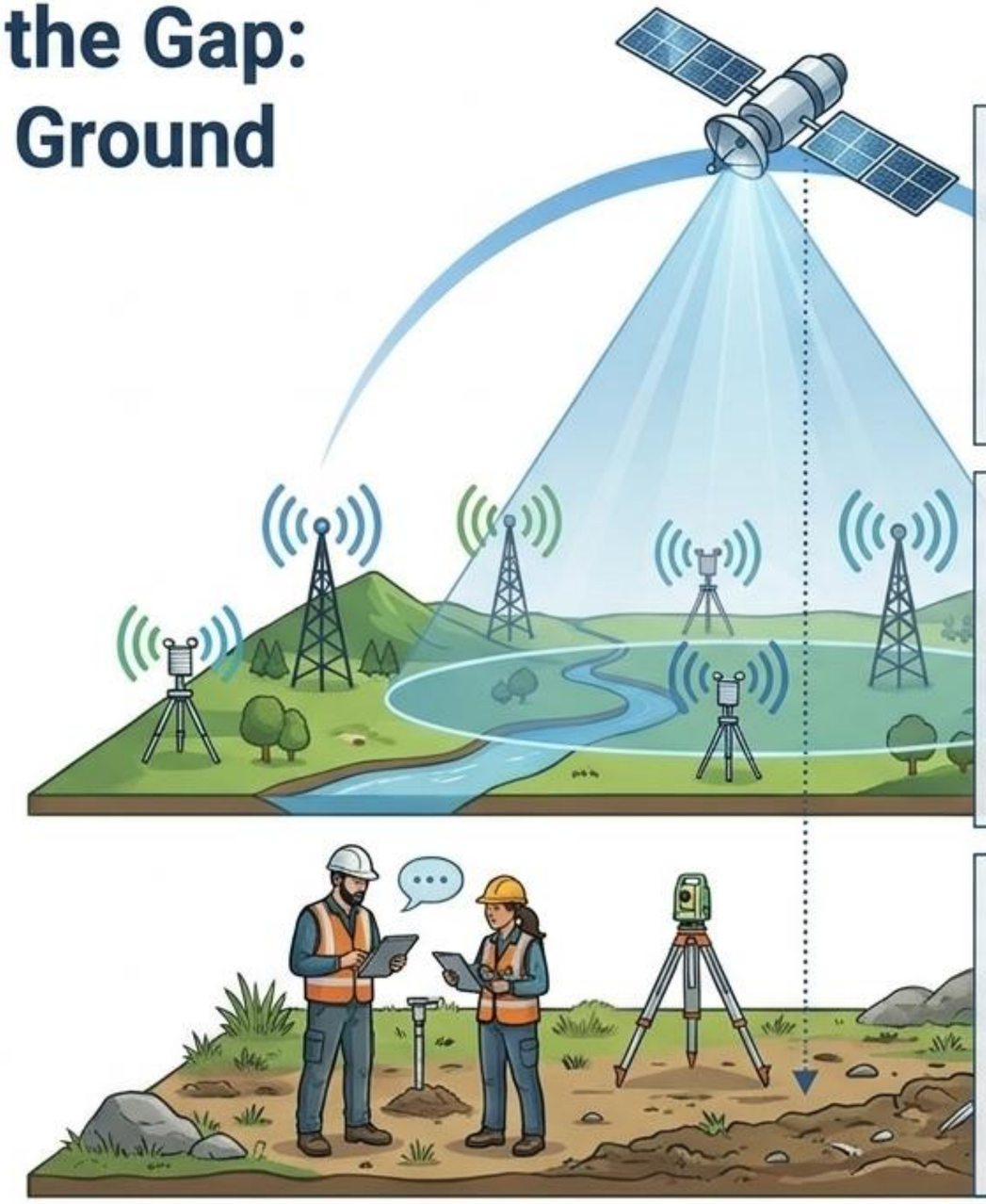


The Multi-Sensor Arsenal: A Capabilities Matrix

Optical (Sentinel-2, Landsat)	**Strengths:** High spatial/spectral resolution, intuitive visual context.	**Blind Spots:** Blocked by clouds and darkness.	**Key Products:** Land cover, vegetation monitoring (NDVI).
SAR (Sentinel-1, ALOS)	**Strengths:** All-weather, day/night imaging, penetrates vegetation (L-band).	**Blind Spots:** Difficult interpretation, high noise/speckle.	**Key Products:** Surface structure, moisture, deformation mapping.
DEM (TanDEM-X, Copernicus)	**Strengths:** 3D structural elevation, highly consistent.	**Blind Spots:** Static snapshot, doesn't capture rapid surface changes.	**Key Products:** Slope, aspect, hydrology modeling.
Climate (ERA5, CHIRPS)	**Strengths:** Global atmospheric and precipitation reanalysis, high temporal frequency.	**Blind Spots:** Very coarse spatial resolution (~31km or 5km).	**Key Products:** Temperature anomalies, drought/flood indicators.

Bridging the Gap: Space to Ground

ASEAN
GEOAI FUSION 2026



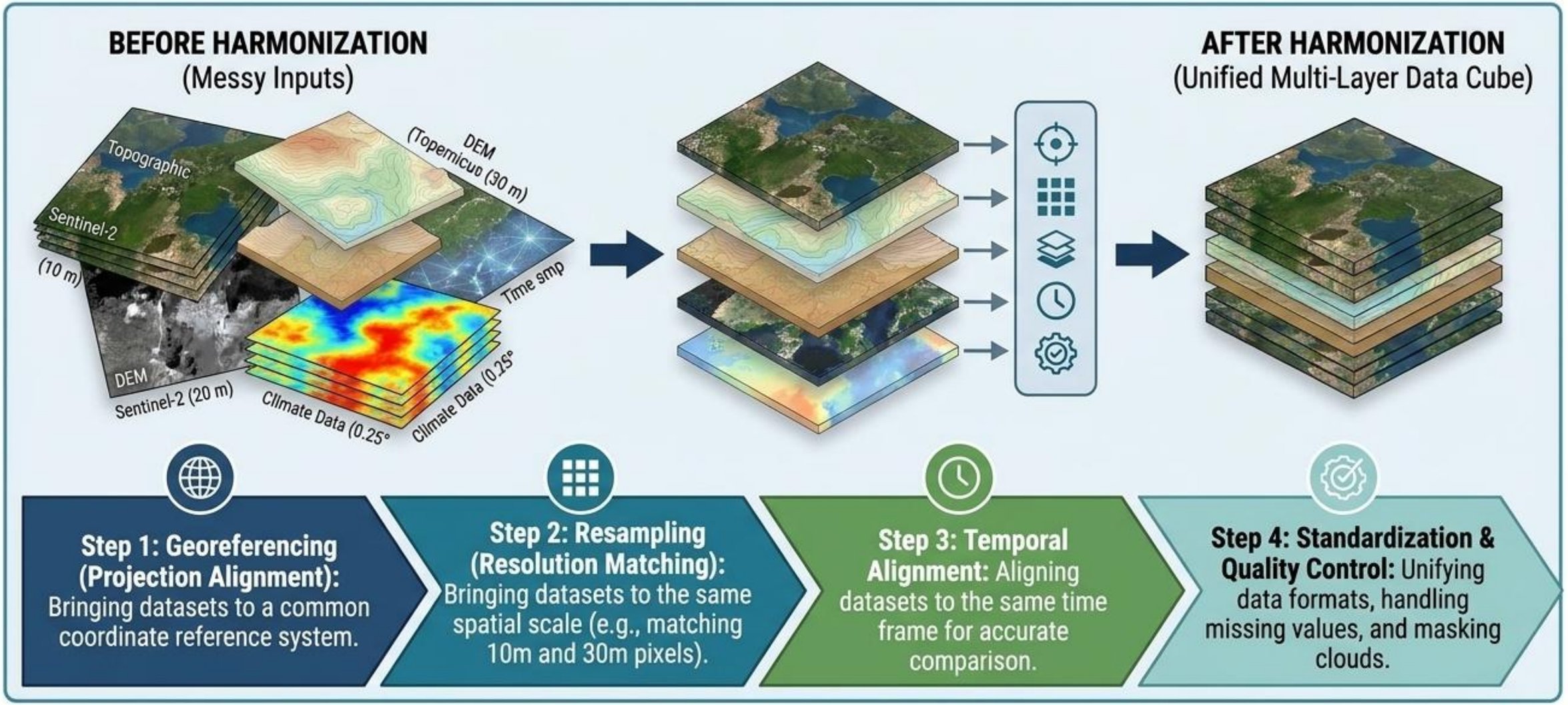
Satellite Imagery (Broad Context):
Predictable revisit schedules capturing vast areas at once. Establishes broad spatial context and historical baselines, though resolution and frequency are fixed by orbits.

Sensor Networks (Rapid Change):
Ground-based IoT (weather stations, air-quality monitors) report continuously from fixed points. Catches rapid, short-lived events that fixed-orbit satellites miss, bridging temporal gaps.

Ground Truth (The Anchor):
Field surveys and verified observations. Serves as the ultimate reference for calibration. Without this, systematic model errors go unnoticed and degrade all downstream intelligence.

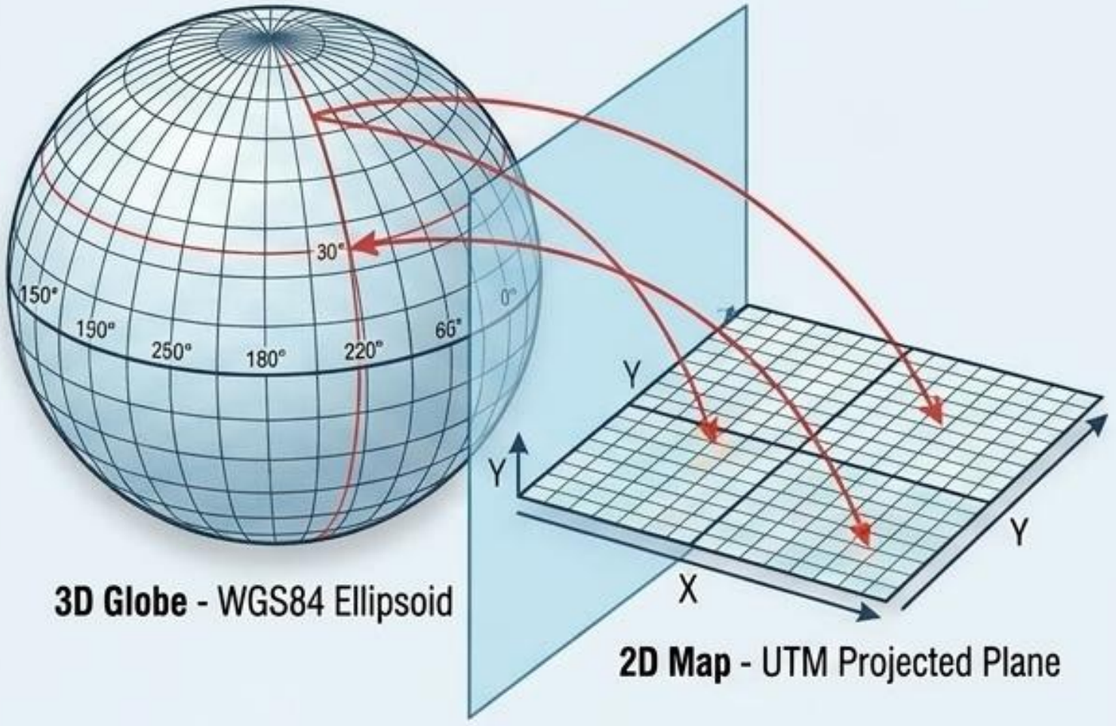
The Foundation of Fusion: Data Harmonization

The Principle: Datasets cannot be fused until they share a single spatial and temporal reality.



Spatial Harmonization: Coordinate Systems and Resampling

Coordinate Reference Systems (CRS)

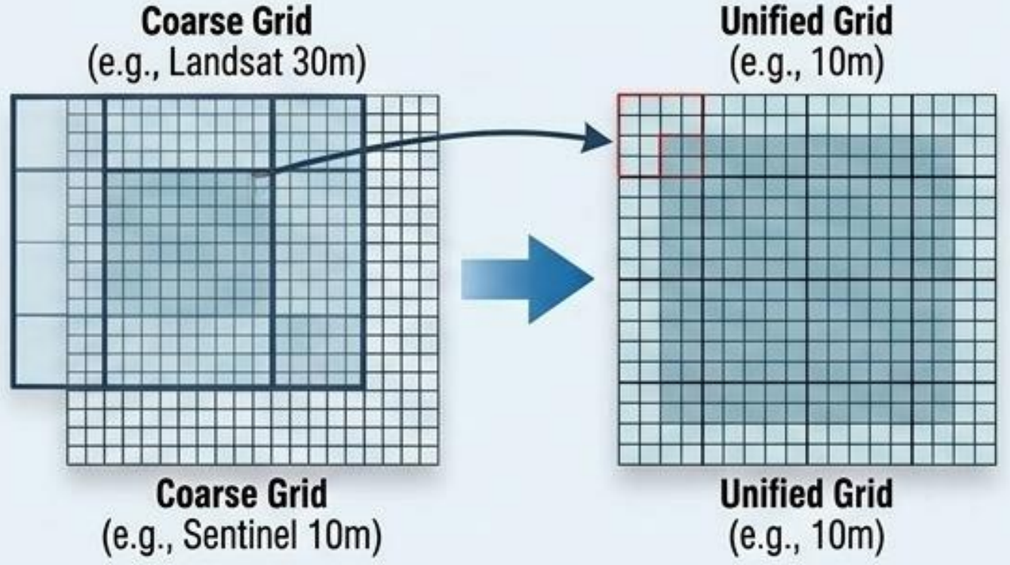


3D Globe - WGS84 Ellipsoid

2D Map - UTM Projected Plane

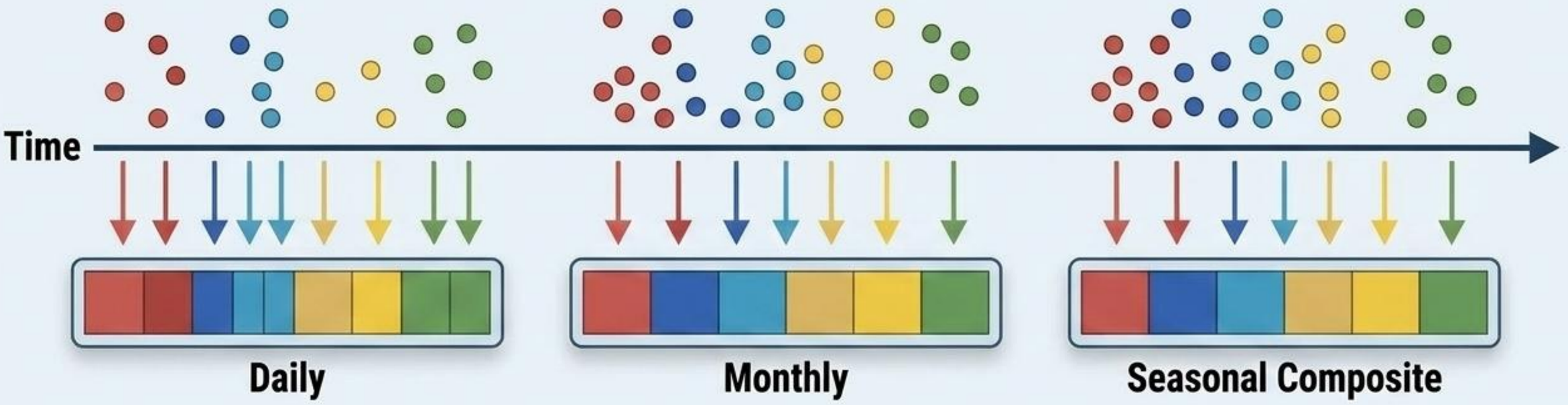
- A dataset in WGS84 (geographic/ellipsoid) and one in a local UTM zone (projected/planar) will not align.
- Subtle georeferencing errors can shift entire fused layers by hundreds of meters, invalidating all subsequent ML analysis.

The Resampling Dilemma



- Sensors capture data at different resolutions (e.g., 10m Sentinel vs 30m Landsat).
- **Nearest Neighbor**: Fastest, preserves original values, best for categorical data (land cover).
- **Bilinear/Cubic**: Smoother, alters values, best for continuous data (elevation, temperature).
- **Rule of Thumb**: Aggregating finer data to a coarser resolution preserves statistical validity better than the reverse.

Temporal Harmonization: Synchronizing the Clocks



The Gap Problem

Sources are rarely captured at the exact same moment. For fast-changing conditions vr. conditions (wildfires, floods), even a few hours of mismatch meaningfully distorts fusion outputs.

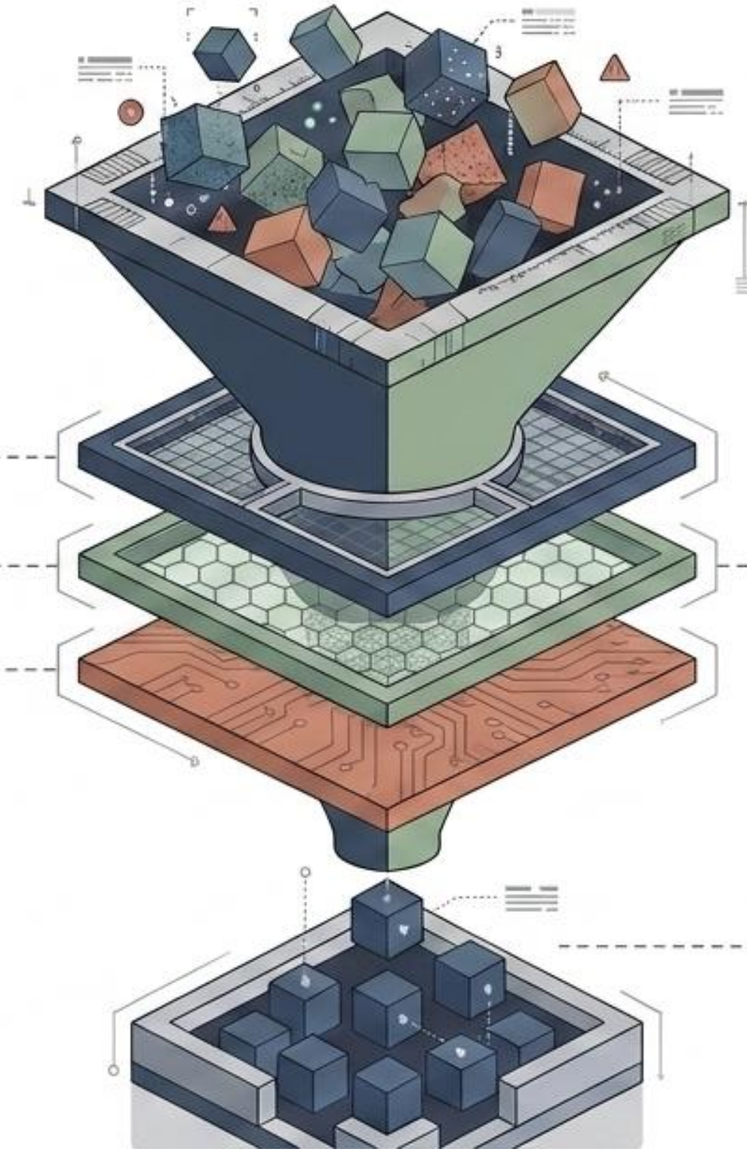
Alignment Strategies

- **Nearest Available:** Selecting the closest temporal acquisition.
- **Interpolation:** Mathematically bridging the gap between two time points.
- **Temporal Aggregation:** Creating composites (Daily, Monthly, Seasonal) to condense vast observation volumes, reduce noise (cloud cover), and reveal long-term seasonal dynamics.

ASEAN
GEOAI FUSION 2026

Data Quality Guardrails

Fusing poor-quality data amplifies errors rather than reducing them.



Accuracy Checks
Comparing dataset values against independent reference points (field surveys). Catching invisible biases before they propagate.

Completeness Management
Explicit strategies for missing data (e.g., cloud-obscured pixels). Options include exclusion, neighbor interpolation, or secondary source substitution.

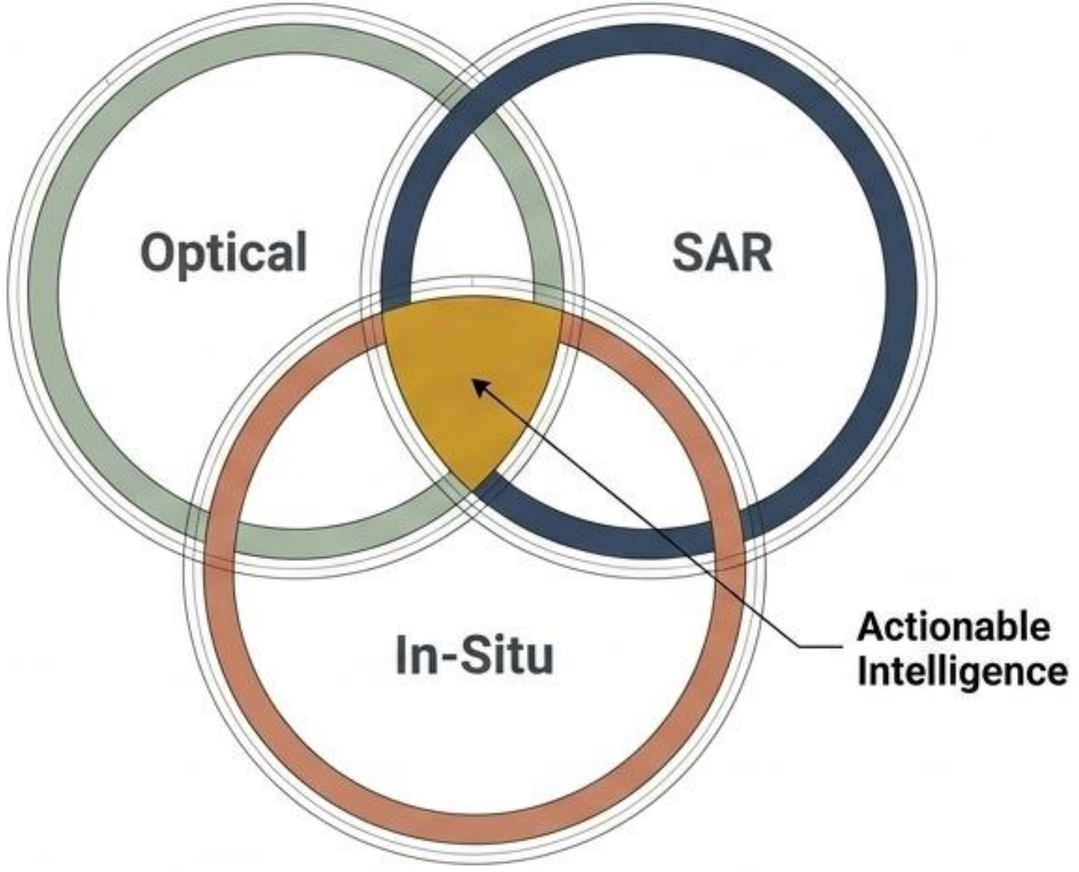
Consistency Resolution
Establishing rules for when sources disagree on the same location/time (e.g., trusting higher-resolution sources or flagging for manual review).

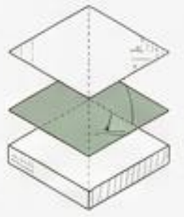
Accuracy Checks
Explicit strategies for ! missing data (e.g., cloud-obscured pixels). Options include exclusion, neighbor neighbor interpolation, or secondary source substitution.

AI-Ready Input

The Alchemy of Data Fusion

Combining sources to produce a reality more accurate, complete, and reliable than any single source.



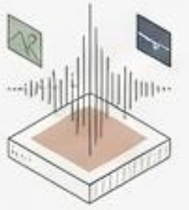


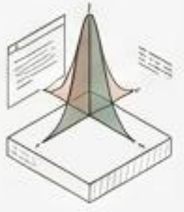
Eradicating Blind Spots

Optical imagery misses cloud-covered zones; radar struggles with fine detail. Fusion pieces together the fragments.

Canceling Noise

Redundancy across different sensor modalities naturally filters out random anomalies and sensor-specific artifacts.





Creating Statistical Reliability

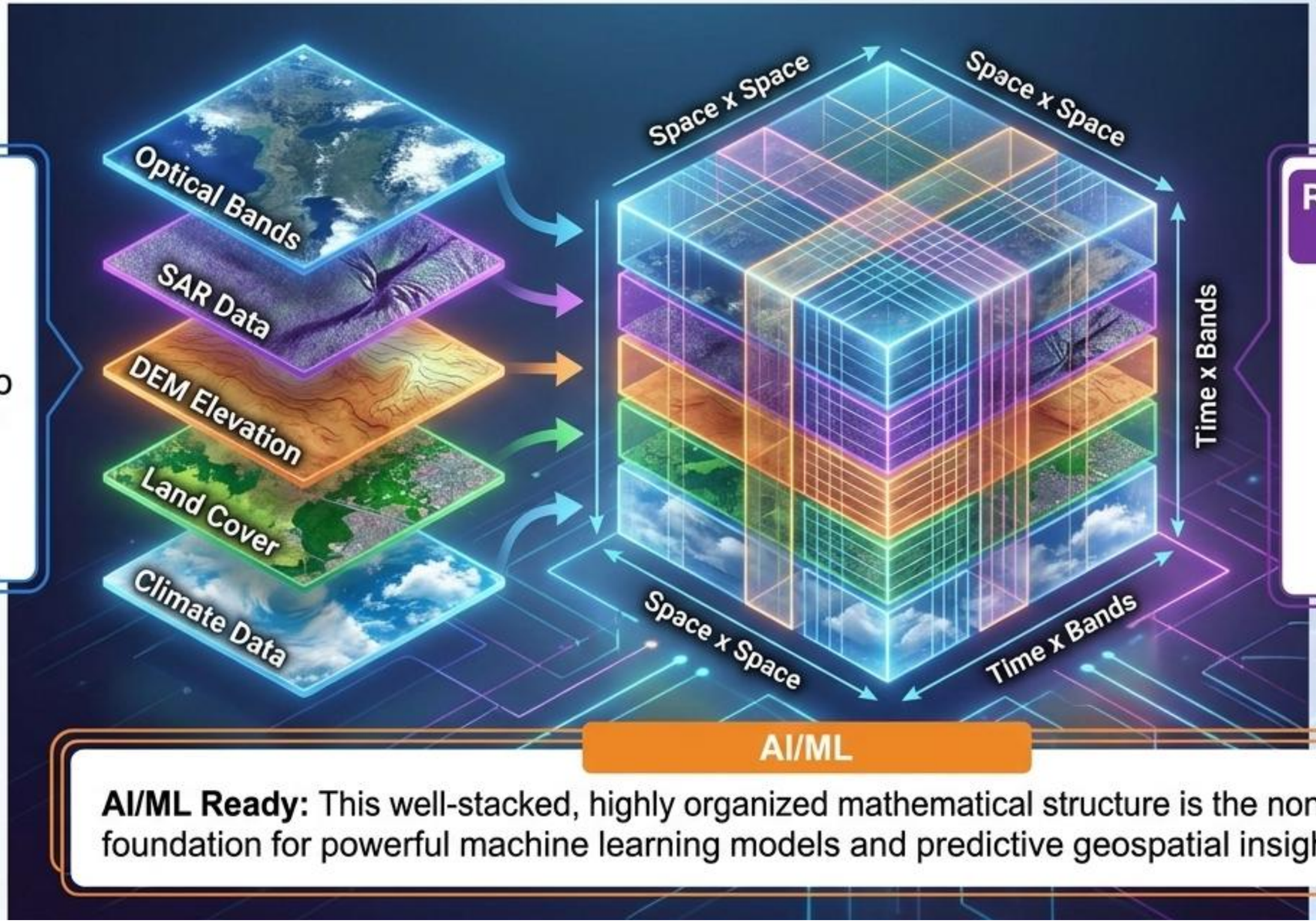
A combined estimate significantly narrows the confidence interval, critical when a missed signal carries a heavy operational cost.

Pre-Fusion Synthesis: Building the 4D Feature Cube

ASEAN GEOAI FUSION 2026

The Goal:

Transform disparate observations into structured data cubes (Space x Space x Time x Bands)



Rich Information Content:

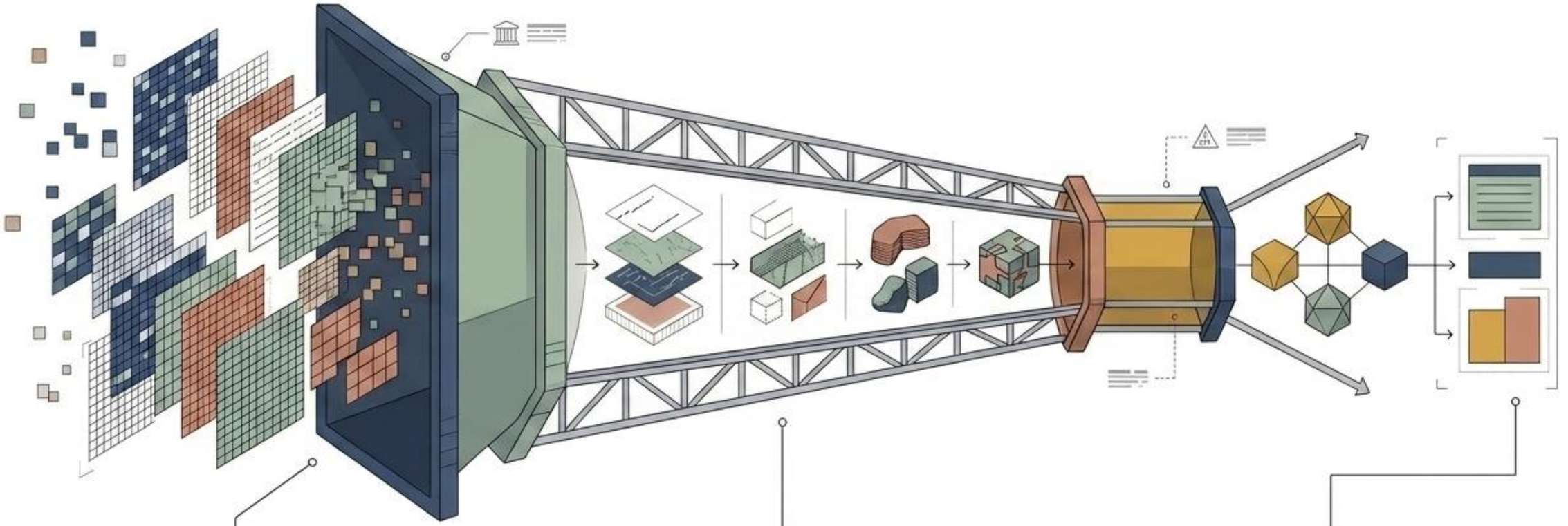
Perfect spatial-temporal alignment integrates multi-source intelligence efficiently

AI/ML

AI/ML Ready: This well-stacked, highly organized mathematical structure is the non-negotiable foundation for powerful machine learning models and predictive geospatial insight.

ASEAN
GEOAI FUSION 2026

The Three Levels of Fusion



Level 1: Pixel-Level Fusion (Raw)

Merging raw sensor values directly (e.g., stacking spectral bands from two different satellite passes before any interpretation occurs).

Level 2: Feature-Level Fusion (Derived)

Combining derived attributes—edges, textures, or objects—after each source has been independently processed and categorized.

Level 3: Decision-Level Fusion (Contextual)

Merging the final outputs of separate classifiers or models. The most robust choice when source data formats differ substantially.

The Algorithmic Engine: How Models Fuse Data



Statistical Methods
(Interpretable & Reliable)

- Weighted averaging combines sources based on reliability.
- Kalman filtering updates estimates dynamically for streaming sensors.
- Bayesian inference formally propagates uncertainty to output a defensible confidence range.



Machine Learning
(Complex & Data-Hungry)

- Neural networks learn complex, nonlinear fusion rules directly from the data without hand-coding.
- Requires large labeled datasets and heavy compute, but outperforms simple statistics when well-trained.



Hybrid Approaches
(The Industry Standard)

- Combines physical, rule-based logic with learned ML models.
- Rules catch obvious physics violations; ML captures highly complex patterns.

ASEAN
GEOAI FUSION 2026

The Multi-Sensor Fusion Framework

Optical + SAR

Combines the spectral richness of optical with the all-weather capability of radar.

Output: Uninterrupted land cover, wetland, and biomass mapping regardless of clouds.

EO Data + Climate Variables

Integrates physical earth observation with meteorology.

Output: Drought, flood, and vegetation stress assessment.

SENSOR PAIRING	FUSION PROCESS	ENHANCED OUTPUT
Optical + SAR	Fusion	All-Weather Land Cover & Biomass Map
Optical + DEM	Fusion	High-Fidelity Topographic Analysis
EO Data + Climate Variables	Fusion	Drought, Flood, & Vegetation Stress Assessment
EO Data + Socioeconomic Data	Fusion	Critical Exposure, Impact, & Disaster Sustainability Assessment

Optical + DEM

Adds elevation context to flat imagery.

Output: High-fidelity topographic and landscape analysis.

EO Data + Socioeconomic Data

Merges the physical environment with human populations and infrastructure.

Output: Critical exposure, impact, and disaster sustainability assessments.



Foundations of Reliable GeoAI

1. **DATA QUALITY DRIVES AI QUALITY**
   Accurate, complete, and consistent data is the inescapable foundation of trustworthy models.

2. **INTEGRATION CANCELS UNCERTAINTY**
   Fusing diverse sensors reduces blind spots and vastly improves real-world robustness.

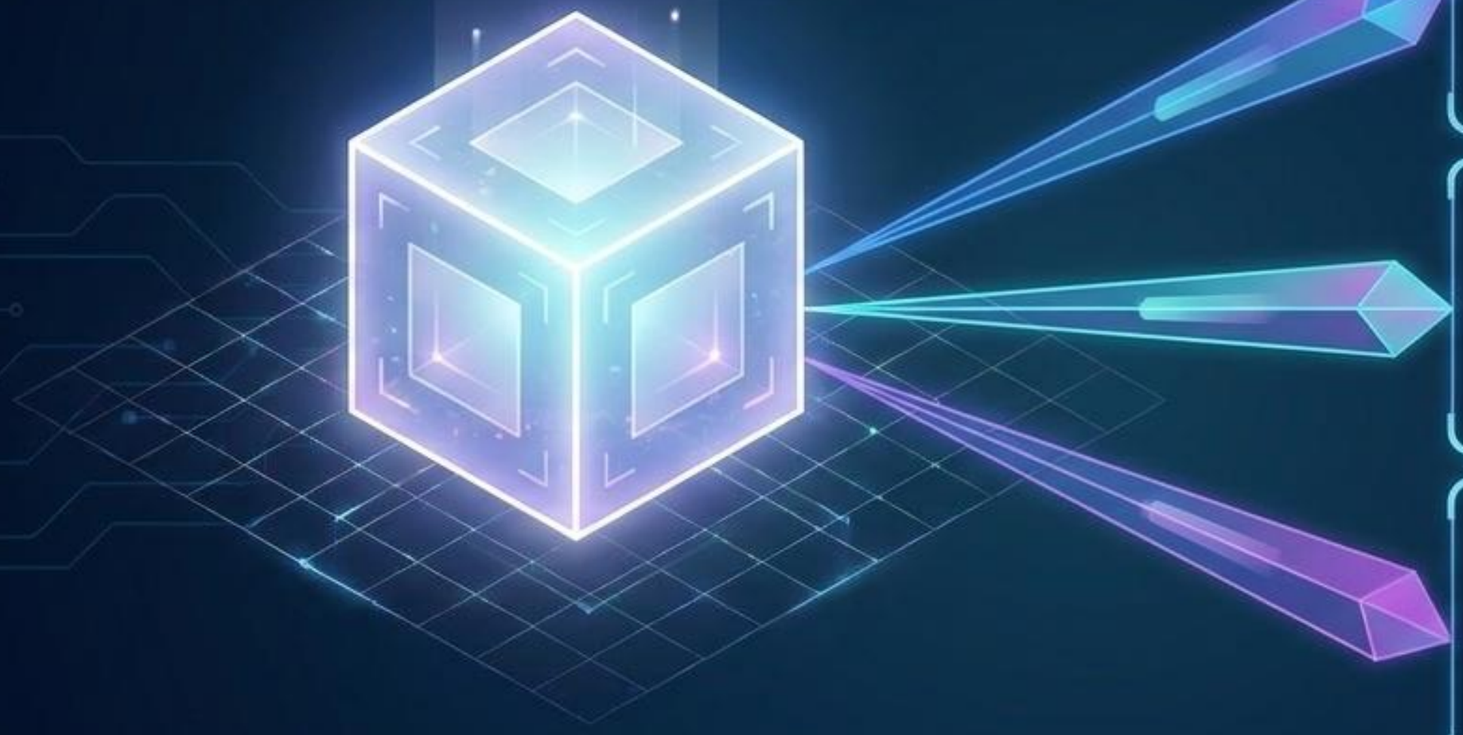
3. **FEATURES DETERMINE PERFORMANCE**
   Well-engineered, domain-relevant features are the key to unlocking true model power.

4. **TIME-SERIES ADD INTELLIGENCE**
   Temporal patterns transition models from static maps to predictive, early-warning forecasts.

5. **PREPARATION IS EVERYTHING**
   Thorough spatial alignment, gap-filling, and validation are what ensure real-world operational impact.

ASEAN
GEOAI FUSION 2026

The Final Output: Operational Intelligence



Disaster Response

Merging satellite wide-views, drone reconnaissance, and IoT ground sensors allows agencies to assess infrastructure damage in hours, not days.

Infrastructure & Telecommunications

Fusing live network KPIs with geospatial and demographic layers enables pinpoint optimization of coverage and resource deployment.

Environmental Protection

Combining weather, soil, and multispectral satellite data provides an undeniable statistical track record of deforestation and water quality degradation.

Conclusion: Fused geospatial intelligence is fundamentally greater than the sum of its raw parts. It is the definitive framework for complex decision-making.